

Advantage: Random forest models specifically excel at reducing overfitting without increasing bias. A single decision tree is prone to high variance but by using bootstrap aggregating, the random forest model averages these trees to create a more stable and generalized boundary (Lecture 7). This is vital for predicting the silent middle because this group could be easily lost in the noise of a single tree.

Impact on individuals:

Moderate citizens are less likely to be misclassified due to noise or isolated extreme answers.

Silent-middle respondents are fairly represented rather than being overshadowed by vocal minorities.

This improves perceived fairness for individuals with balanced views.

Impact on organisations:

Policymakers can rely on stable and robust results we get from our model

Decisions are less influenced by noisy or extreme opinions given by people in the survey

Impact on society:

Helps prevent policy outcomes from being dominated by vocal minorities who give extreme opinions

Reduces the risk of polarization driven by extreme viewpoints

Advantage: Another strength of using the random forest model is that it does not require the scaling of data (Lecture 7). Our dataset uses a wide range of scoring systems and different variables that are used by only small groups in the data. Because random forest does not require the normalization of data, unlike for example neural networks, it can handle mixed-type survey data the best.

Impact on individuals:

Because the model does not require scaling, individual responses are not distorted much. Variables that are used by small groups or measured on smaller scales are not artificially downweighted compared to variables with larger numeric ranges.

Impact on organisations:

For organisations and policymakers, the lack of required scaling simplifies data preprocessing and reduces the likelihood of errors. Incorrect normalization choices can unintentionally bias results or amplify certain variables over others. By using a model that is robust to mixed scales, organisations can more confidently analyze complex survey data.

Impact on society: When machine learning models do not rely on scale-sensitive preprocessing, the risk that technical design decisions disproportionately affect certain groups

is reduced. This contributes to responsible and fair use of AI in democratic contexts, ensuring that diverse forms of citizen input are reflected accurately in data-driven decision-making.

Disadvantage:. In lecture 3 of XAI, we see that the random forest model is placed high on the model accuracy scale, but lower on model interpretability. This model is seen as a black box approach which is a risk when trying to explain the citizen participation (silent middle) for a socio-technical system. To increase the interpretability we have to add another layer of complexity by introducing methods like SHAP to understand why a citizen was labeled “silent middle”. This increased complexity is a risk because it could create misunderstanding and potential errors with policy makers

Impact on individuals:

Individuals may not understand why they are grouped as part of the silent middle. This is because Random Forest predictions result from the aggregation of many decision trees rather than from simple, human-readable rules. When explanations rely on SHAP values, they provide statistical approximations rather than proper intuitive reasoning. As a result, citizens may perceive decisions about their participation as opaque or unfair

Impact on organisations:

For organisations and policymakers, the reliance on post-hoc explanation methods increases the risk of misinterpretation. SHAP values indicate correlations and contribution strength, but they can easily be mistaken for causal explanation, because policymakers may lack expertise in machine learning, they may over-trust these explanations to draw incorrect conclusions.

Impact on Society:

At the societal level, the use of such models in democratic contexts threatens transparency and accountability. Over time, repeated use of opaque systems may undermine democratic legitimacy, even if the model achieve high predictive performance.

Disadvantage:. Our model identifies the grade (Grade the participant gave to the consultation) as a top predictor for the silent middle, but in lecture 1 it is said that machine learning models: "operate mostly on associations and are agnostic about causal direction". An analytical limitation of our random forest model is that it cannot prove that a high grade causes someone to be in the silent middle; it only shows they are correlated. This is a significant limitation when trying to design interventions to engage those citizens.

Impact on individuals:

Individuals may be incorrectly targeted by interventions designed to increase satisfaction, under the assumption that higher grades will lead to silent-middle behavior. Because the model captures correlation rather than causation, such interventions may fail to address individuals' actual preferences.

Impact on organisations: For organisations and policymakers, relying on correlational insights can lead to poorly designed interventions. If decision-makers interpret grade as evidence of a causal

relationship, resources may be allocated to increasing satisfaction without improving participation quality.

Impact on Society: Over time, reliance on such models can undermine confidence in algorithmic based decision-making within democratic systems.